

ES 671: Mechanics of Composite Materials

End-Semester Project Report

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I. FAILURE CRITERIA FOR AN ORTHOTROPIC ANGLE LAMINA

A. Problem Statement

An orthotropic angle lamina making an angle θ with the longitudinal direction is subjected to a uniaxial stress σ_x . The normalised failure stress σ_x/F_{1t} is plotted as a function of θ for four failure criteria: Maximum Stress, Maximum Strain, Tsai-Hill, and Tsai-Wu.

B. Theory

1) *Stress Transformation*: The on-axis stresses ($\sigma_1, \sigma_2, \tau_{12}$) are obtained from the off-axis stress σ_x via:

$$\sigma_1 = \sigma_x \cos^2 \theta, \quad (1)$$

$$\sigma_2 = \sigma_x \sin^2 \theta, \quad (2)$$

$$\tau_{12} = -\sigma_x \sin \theta \cos \theta. \quad (3)$$

2) *Maximum Stress Criterion*: Failure is predicted when any on-axis stress component reaches its respective strength:

$$\frac{\sigma_x}{F_{1t}} = \min \left(\frac{F_{1t}}{\cos^2 \theta}, \frac{F_{2t}}{\sin^2 \theta}, \frac{F_6}{|\sin \theta \cos \theta|} \right)^{-1}. \quad (4)$$

3) *Maximum Strain Criterion*: The compliance matrix is used to compute strains:

$$\varepsilon_1 = \frac{\sigma_1}{E_1} - \frac{\nu_{12}\sigma_2}{E_1}, \quad (5)$$

$$\varepsilon_2 = \frac{\sigma_2}{E_2} - \frac{\nu_{21}\sigma_1}{E_2}, \quad (6)$$

$$\gamma_{12} = \frac{\tau_{12}}{G_{12}}. \quad (7)$$

Failure occurs when any strain reaches its ultimate value ($\varepsilon_{1u} = F_{1t}/E_1$, etc.).

4) *Tsai-Hill Criterion*:

$$\left(\frac{\sigma_1}{X_1} \right)^2 - \frac{\sigma_1\sigma_2}{X_1^2} + \left(\frac{\sigma_2}{X_2} \right)^2 + \left(\frac{\tau_{12}}{S} \right)^2 = 1 \quad (8)$$

where $X_1 = F_{1t}$ if $\sigma_1 > 0$, else $|F_{1c}|$; and $X_2 = F_{2t}$ if $\sigma_2 > 0$, else $|F_{2c}|$, and $S = F_6$

5) *Tsai-Wu Criterion*:

$$F_1^* \sigma_1 + F_2^* \sigma_2 + F_{11}^* \sigma_1^2 + F_{22}^* \sigma_2^2 + F_{66}^* \tau_{12}^2 + 2F_{12}^* \sigma_1 \sigma_2 = 1, \quad (9)$$

where the strength parameters are defined as:

$$F_1 = \frac{1}{F_{1t}} + \frac{1}{F_{1c}}, \quad F_2 = \frac{1}{F_{2t}} + \frac{1}{F_{2c}} \quad (10)$$

$$F_{11} = \frac{-1}{F_{1t}F_{1c}}, \quad F_{22} = \frac{-1}{F_{2t}F_{2c}}, \quad F_{66} = \frac{1}{S^2} \quad (11)$$

$$F_{12} = -\frac{1}{2} \sqrt{F_{11}F_{22}} \quad (12)$$

C. Results

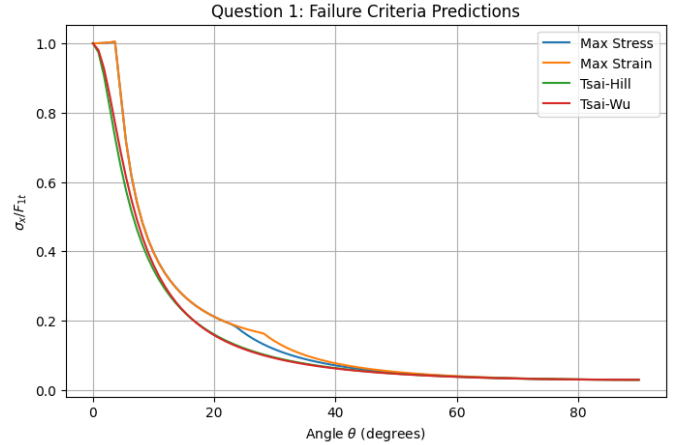


Fig. 1. Normalised failure stress σ_x/F_{1t} vs. lamina angle θ for the four failure criteria.

D. Observations

- All four criteria agree closely near $\theta = 0^\circ$ and $\theta = 90^\circ$, where fibre-direction and transverse strengths respectively dominate.
- The shear-dominated region ($\theta \approx 45^\circ$) shows the greatest spread between criteria; Maximum Stress - most conservative there, Tsai-Wu - least conservative owing to its interaction term.
- Tsai-Hill and Tsai-Wu produce smooth, differentiable curves, making them preferable for gradient-based optimisation.

- The Maximum Strain criterion deviates slightly from the Maximum Stress criterion because of the Poisson coupling between ε_1 and ε_2 .
- For small angles ($\theta < 10^\circ$) fibre tensile failure governs all criteria; for large angles ($\theta > 75^\circ$) transverse tensile failure governs.

II. SHEAR STRAIN IN LAMINATES UNDER COMBINED LOADING

A. Problem Statement

A 1 mm thick symmetric laminate is loaded by N_x and N_{xy} with $N_x = 2N_{xy}$. The shear strain γ_{xy} is determined using Classical Lamination Theory (CLT) for nine different layups, and the results are normalised as γ_{xy}/N (where $N \equiv N_{xy}$).

B. Theory

For a symmetric laminate the mid-plane strains are obtained from the $\mathbf{A}$ matrix:

$$\begin{Bmatrix} \varepsilon_x^0 \\ \varepsilon_y^0 \\ \gamma_{xy}^0 \end{Bmatrix} = \mathbf{A}^{-1} \begin{Bmatrix} N_x \\ N_y \\ N_{xy} \end{Bmatrix}. \quad (13)$$

With $N_y = 0$ and $N_x = 2N_{xy}$:

$$\gamma_{xy}^0 = (a_{31} \cdot 2 + a_{33}) N_{xy}, \quad (14)$$

where a_{ij} are the components of $\mathbf{A}^{-1}$.

C. Results

TABLE I
SHEAR STRAIN NORMALISED BY N_{xy} FOR EACH LAYUP

Case	Layup	γ_{xy}/N (1/(Pa·m))
1	[0]	2.415459×10^{-7}
2	[0/15] _s	1.766154×10^{-7}
3	[0/25] _s	1.393208×10^{-7}
4	[0/30] _s	1.292271×10^{-7}
5	[0/45] _s	1.355963×10^{-7}
6	[0@t/25@1.5t] _s	1.266975×10^{-7}
7	[0@t/25@2t] _s	1.188410×10^{-7}
8	[0@t/25@2.5t] _s	1.134134×10^{-7}
9	[0@t/30@2.5t] _s	1.016437×10^{-7}

III. STRESS DISTRIBUTION IN A [30/75/30] LAMINATE

A. Problem Statement

A [30/75/30] laminate with each ply of thickness 0.75 mm is analysed under an applied bending moment converted to standard SI units as $M_x = 100$ N·m/m. The $\mathbf{A}$, $\mathbf{B}$, and $\mathbf{D}$ matrices are computed, and the through-thickness stress distributions σ_x , σ_y , and τ_{xy} are plotted.

B. Theory

For a general laminate the mid-plane strains and curvatures satisfy:

$$\begin{Bmatrix} \mathbf{N} \\ \mathbf{M} \end{Bmatrix} = \begin{bmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{B} & \mathbf{D} \end{bmatrix} \begin{Bmatrix} \varepsilon^0 \\ \kappa \end{Bmatrix}. \quad (15)$$

$$A_{ij} = \sum_{k=1}^n \bar{Q}_{ij}(z_k - z_{k-1}) \quad (16)$$

$$B_{ij} = \frac{1}{2} \sum_{k=1}^n \bar{Q}_{ij}(z_k^2 - z_{k-1}^2) \quad (17)$$

$$D_{ij} = \frac{1}{3} \sum_{k=1}^n \bar{Q}_{ij}(z_k^3 - z_{k-1}^3) \quad (18)$$

With $\mathbf{N} = \mathbf{0}$ and only M_x applied, the system reduces to a 6×6 linear solve. The stress in the k -th ply is:

$$\begin{Bmatrix} \sigma_x \\ \sigma_y \\ \tau_{xy} \end{Bmatrix}_k = \bar{\mathbf{Q}}_k (\varepsilon^0 + z \kappa), \quad z \in [z_{k-1}, z_k]. \quad (19)$$

C. ABD Matrices

The computed stiffness matrices in standard SI units are:

$$\mathbf{A} = 10^7 \times \begin{bmatrix} 4.631 & 1.365 & 1.504 \\ 1.365 & 4.322 & 1.072 \\ 1.504 & 1.072 & 1.805 \end{bmatrix} \text{ (N/m)} \quad (20)$$

$$\mathbf{B} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \text{ (N)} \quad (21)$$

$$\mathbf{D} = \begin{bmatrix} 24.51 & 6.69 & 8.76 \\ 6.69 & 11.38 & 3.69 \\ 8.76 & 3.69 & 8.55 \end{bmatrix} \text{ (N·m)} \quad (22)$$

D. Results

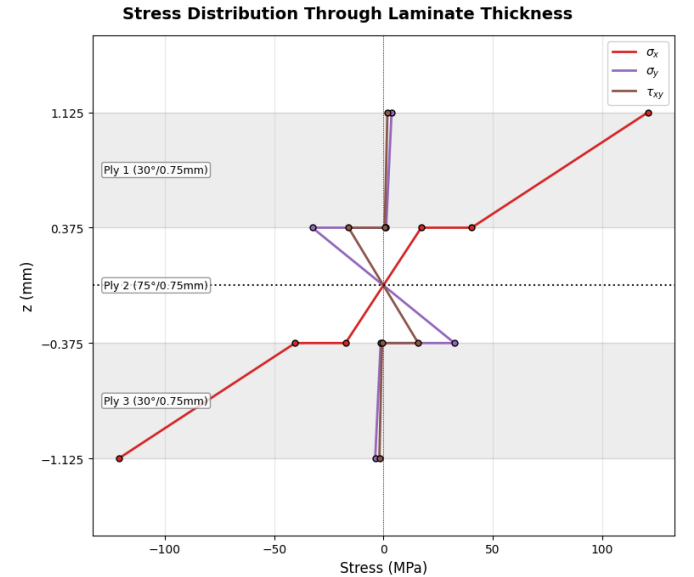


Fig. 2. Through-thickness stress distributions σ_x , σ_y , and τ_{xy} in the [30/75/30] laminate under $M_x = 100$ N·m/m.

E. Stress Summary at Ply Interfaces

TABLE II
STRESS VALUES (MPa) AT THE TOP AND BOTTOM FACES OF EACH PLY

Layer	Position	σ_x	σ_y	τ_{xy}
Ply 1 (30°)	Top	121.08	3.75	1.84
	Bottom	40.36	1.25	0.61
Ply 2 (75°)	Top	17.33	-32.53	-15.94
	Bottom	-17.33	32.53	15.94
Ply 3 (30°)	Top	-40.36	-1.25	-0.61
	Bottom	-121.08	-3.75	-1.84

IV. BENDING-EXTENSION COUPLING IN SYMMETRIC ANGLE-PLY LAMINATES

A. Problem Statement

A regular symmetric angle-ply laminate $[+\theta/-\theta/\theta/\dots]$ with total thickness $h = 3$ mm and equal-thickness plies is studied. The normalised extension-shear coupling $A_{16}/(A_{11}h)$ and bending anisotropy $-D_{16}/D_{11}$ are plotted vs. θ for $n = 3, 5, 7, 9, 11$ layers.

B. Theory

For a regular symmetric angle-ply laminate with finite layers, A_{16} is non-zero because the $+\theta$ and $-\theta$ plies do not appear in equal numbers, causing incomplete cancellation of shear-extension coupling. As $n \rightarrow \infty$, the laminate approaches a truly balanced layup and $A_{16} \rightarrow 0$. Similarly, D_{16} is non-zero for finite layers but diminishes with increasing n , making the laminate progressively more orthotropic in both extension and bending.

C. Results

The question initially asked to plot $B_{16}/(A_{11}h)$ vs. θ , but that is just a straight line at 0 since matrix B is zero. So for clarity and insight, $A_{16}/(A_{11}h)$ vs. θ was plotted.

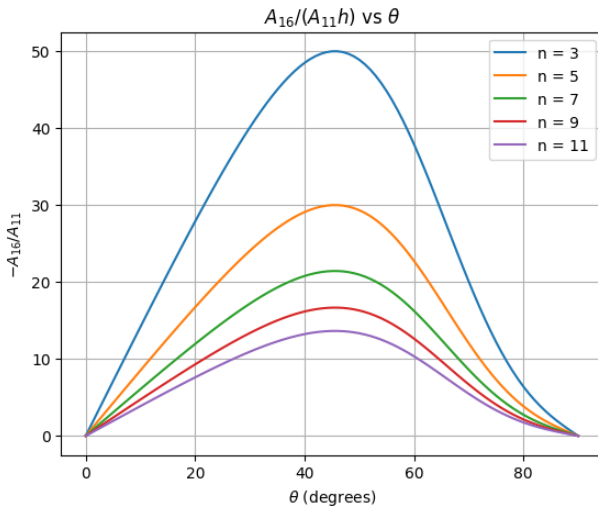


Fig. 3. Extension-Shear Coupling $A_{16}/(A_{11}h)$ vs. θ

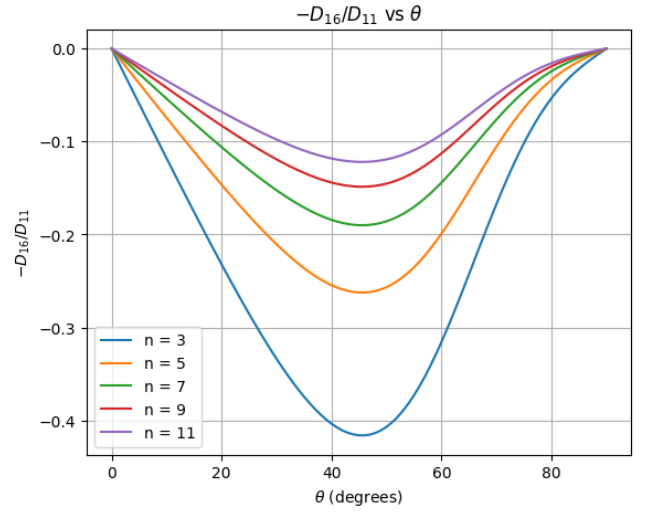


Fig. 4. Bending-Twist Coupling $-D_{16}/D_{11}$ vs. θ

D. Maximum Coupling Angle

$$B_{16}/(A_{11}h) \approx 0^\circ \quad \forall \theta$$

$$A_{16}/(A_{11}h) \approx 46^\circ \quad \forall \theta$$

E. Observations

- The extension-shear coupling coefficient $A_{16}/(A_{11}h)$ decreases significantly as the number of layers n increases. For $n = 3$, the peak value reaches ≈ 50 , while for $n = 11$ it reduces to ≈ 13 , more layers $\implies$ more balanced.
- The bending anisotropy $-D_{16}/D_{11}$ depends on the ply angle, with its maximum effect occurring near $\theta \approx 46^\circ$.
- As the number of alternating layers increases ($n \rightarrow 11$), the magnitude of $-D_{16}/D_{11}$ steadily decreases. This demonstrates that using many thin, alternating angle plies effectively "washes out" the bending-twisting coupling.
- At $\theta = 0^\circ$ and $\theta = 90^\circ$ both coefficients vanish entirely because all plies align with the primary structural axes.

CONCLUSION

This study applied Classical Lamination Theory and standard failure criteria to four benchmark problems. Key findings: (i) failure criteria diverge most in the shear-dominated range $30^\circ \lesssim \theta \lesssim 60^\circ$; (ii) shear strain under combined loading is highly sensitive to off-axis ply proportion; (iii) symmetric laminates have zero extension-bending coupling but retain stress jumps at interfaces due to discontinuous stiffness; and (iv) bending-twisting coupling diminishes rapidly with increasing ply count.

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